

Can Chance Make a Streak?

AP Statistics · Topic 4.1 (CED 2.3) · B: 2026-10-19 · E: 2026-11-02 · TEACHER KEY

CED TETHER

- **VAR-1.F** Identify questions suggested by patterns in data. [**Skill 1.A**]
- **VAR-1.F.1** Patterns in data do not necessarily mean that variation is not random.

Essential question: How can simulation distinguish a surprising-looking random pattern from evidence against a chance model?

DOK-3 skill: 4.A · **FRQ pattern:** is-this-pattern-random-design-the-simulation

DOK rationale: Part (c) weighs a striking pattern against simulated evidence, then coordinates a fair-coin model, an at-least-as-extreme statistic, repeated trials, and a result that would challenge the model.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) **DOK: 1**
→ 3 **Minutes: 45**

Teacher does

- Board slide up at the bell. Let the six-head streak feel surprising without endorsing either claim.
- Begin the combined 4.1–4.2 video follow-along at minute 5; leave first takes ungraded.
- Return to the board slide at minute 33. Circulate and press students to match a trial to all 10 flips.
- Collect at the door. Score part (c) only, E/P/I.

Students do

- Commit to one claim and cite the sequence or dotplot before the lesson.
- Complete the follow-along about random patterns and simulation.
- Finish (a)–(c), revise the first take in the exit line, and turn in the sheet.

Questions to ask

- What feature of HHHHHHTTTT are you measuring: total heads or consecutive outcomes?
- Why must a trial contain 10 flips rather than only the six-flip streak?
- Should a run of 7 count as at least as surprising as the observed run?
- What simulated proportion would make this result genuinely unusual under fairness?

Adult role

Solo teacher. Circulate 5 pairs. Ask students to name model, trial, statistic, event, and repetition count before accepting a conclusion about the coin.

[WARN] WATCH FOR

- Treating an eye-catching streak as impossible under randomness.
- Counting total heads rather than longest run without explaining a different statistic.

- Claiming that a 7 percent simulation estimate proves the coin fair.

SCORING PART (c) — E / P / I

Expected elements

- **chooses-simulation-claim** (required) — Chooses Claim B and cites that 7 of 100 fair-coin trials produced a longest run of at least 6
- **defines-matched-trial** (required) — Defines one trial as 10 independent equally likely fair-coin flips
- **records-extreme-statistic** (required) — Records the longest run and counts trials whose longest run is at least 6
- **uses-many-trials-and-doubt-criterion** (required) — Uses many trials and states that a stable, extremely small successful-trial proportion would count against fairness

E	Chooses Claim B with the 7-of-100 evidence, gives a complete matched simulation with many trials, and identifies a very small stable proportion as evidence against fairness without claiming the current result proves fairness.
P	Makes the supported claim choice and gives most of the simulation, but omits the at-least-6 event, a large repetition count, or a clear result that would challenge fairness.
I	Says the streak cannot occur by chance, simulates a different process, counts total heads instead of longest runs without justification, or gives no evidence-based decision.

Common mistakes

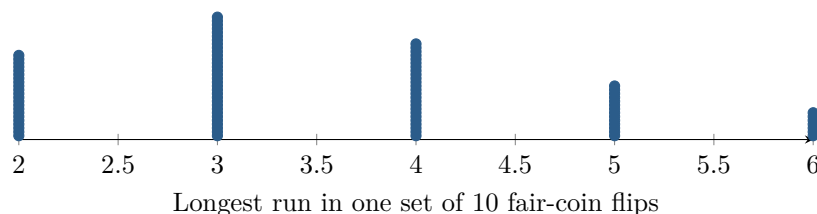
- Treating a run of six as impossible because each individual sequence is unlikely
- Using six flips rather than reproducing all 10 flips in one trial
- Counting only runs of exactly 6 and discarding longer runs
- Claiming that 7 percent proves the coin is fair

[STAR] TODAY'S PROBLEM — annotated

Suppose a coin advertised as fair is flipped 10 times and gives HHHHHHTTTT. To investigate what chance alone could produce, a class simulates 100 sets of 10 independent fair-coin flips. For each set, the class records the longest run of matching outcomes; the dotplot shows the results.

Claim A: “The coin is rigged—six heads in a row cannot be chance.”

Claim B: “Streaks like this can happen by chance; simulation can tell us how unusual this one is.”



FIRST TAKE (5 min)

Which claim seems better supported right now? Cite one detail from the sequence or dotplot.

Key: Not graded. Preserve surprise; the key move is replacing visual intuition with a matched chance model.

Rules callout “MODEL THE PATTERN, NOT THE STORY (keep on the page)” is printed on the student sheet.

DOK 1 (a) State the longest run in the observed sequence and the most common longest-run value among the simulated sets.

Key: The observed sequence has a longest run of 6, the six consecutive heads. In the simulated sets, 3 is the most common longest-run value, appearing 32 times.

DOK 2 (b) Use the dotplot to estimate the probability that 10 fair-coin flips have a longest run at least as long as the observed run. Interpret the estimate in context.

Key: Seven of the 100 simulated sets have a longest run of at least 6, so the estimate is $7/100 = 0.07$. In context, about 7 percent of sets of 10 independent flips from a fair coin produced a run at least as long as the observed run in this simulation.

DOK 3 (c) Which claim is better supported by this evidence? Cite the simulation result that settles your choice. Then describe a larger simulation that would more precisely settle the disagreement: state what one trial is, what to record and count, how many trials to use, and what result would make you doubt that the coin is fair.

Key: Claim B is better supported. The fair-coin simulation produced a run of 6 in 7 of 100 trials, so a streak this long is possible by chance; 100 trials give only a rough estimate and do not prove the coin fair. For a stronger simulation, use independent equally likely H/T outcomes. One trial is 10 flips; record the longest run of either outcome and mark the trial a success when that run is at least 6. Repeat at least 10,000 trials and compute the successful-trial proportion. A stable proportion extremely close to zero—for example, below 1 percent—would make the observed run very unusual under the fair-coin model and provide reason to doubt fairness. A proportion near the displayed 7 percent would not.

EXIT

One thing the video changed about my first take: _____